Appendix

Title: From Technology to Trust: Advancing Human-Centric AI and Digital Sovereignty

The survey instrument included 21 items organized into five latent constructs: Ethical Trust (ET), Functional Trust (FT), Institutional Trust (IT), Citizen Trust (CT), and Service Acceptance (SA). All items are estimated using a five-point Likert scale, ranging from 1 (Strongly Disagree) to 5 (Strongly Agree). The survey form is included in Table 6. We have designed four hypotheses for validation as follows.

H1: Ethical trust positively influences citizen trust.

H2: Functional trust positively influences citizen trust.

H3: Institutional trust positively influences citizen trust.

H4: Citizen trust positively influences service acceptance.

Table 6. Survey form.

Construct	Code	Indicator
Ethical Trust	ET1	Transparency
	ET2	Explainability
	ET3	Accountability
	ET4	Privacy Protection
	ET5	Digital Sovereignty
Functional Trust	FT1	Human Oversight
	FT2	Autonomy
	FT3	Safety
	FT4	Reliability
	FT5	Digital Literacy
Institutional Trust	IT1	Sustainability
	IT2	Governance Transparency
	IT3	Accountability Audits
	IT4	Ethical Alignment
Citizen Trust	CT1	Trustworthiness
	CT2	Confidence
	CT3	Perceived Security
	CT4	Reliability
Service Acceptance	SA1	Intention to Use
	SA2	Continued Usage Intention
	SA3	Recommendation Intention

Cronbach's Alpha is used to assess internal consistency, as shown in Table 7. The outcomes show satisfactory reliability for ethical trust, functional trust, citizen trust, and service acceptance. However, the institutional trust showed a lower alpha value, but the analysis is limited to a small sample size.

Table 7. Reliability analysis,

Elements	Cronbach's Alpha
Ethical Trust	0.856
Functional Trust	0.850
Institutional Trust	0.508
Citizen Trust	0.854
Service Acceptance	0.834

The structural relationships among the constructs are analyzed using regression-based path analysis, as shown in Table 8. The findings indicate that all proposed relationships are positive and statistically meaningful. Among the antecedents of citizen trust, institutional trust ($\beta = 0.496$) demonstrated the strongest influence, followed by ethical trust ($\beta = 0.454$) and functional trust ($\beta = 0.258$).

Table 8. Hypothesis testing outcomes

Hypothesis	Path	β	Decision
H1	Ethical Trust → Citizen Trust	0.454	Supported
H2	Functional Trust → Citizen Trust	0.258	Supported
H3	Institutional Trust → Citizen Trust	0.496	Supported
H4	Citizen Trust → Service Acceptance	0.777	Supported

We performed an R^2 test for the proposed model, as shown in Table 9. The ethical, functional, and institutional trust layers collectively accounted for 66.9% of the variance in citizen trust, demonstrating substantial predictive power. Also, citizen trust explained 58.8% of the variance in service acceptance, indicating that trust is a key determinant of citizens' readiness to adopt AI-enabled services in 5G smart-city environments.

Table 9. Coefficient determinations.

Construct	R^2
Citizen Trust	0.669
Service Acceptance	0.588

The reliability and convergent validity of the proposed model are evaluated using Cronbach's Alpha (CA), Composite Reliability (CR), and Average Variance Extracted (AVE). All constructs accomplished CR values above the recommended threshold of 0.70 and AVE values above 0.50, ensuring satisfactory convergent validity, as shown in Table 10.

Table 10. Reliability and convergent validity.

Construct	CA	CR	AVE
Ethical Trust (ET)	0.856	0.897	0.637
Functional Trust (FT)	0.85	0.892	0.624
Institutional Trust (IT)	0.731	0.832	0.557
Citizen Trust (CT)	0.854	0.901	0.694
Service Acceptance (SA)	0.834	0.889	0.728

We have used bootstrapping with 5,000 subsamples to test the presented hypotheses. The outcomes are shown in Table 11. The outcomes support all proposed hypotheses. Institutional trust exhibits the strongest positive effect on citizen trust ($\beta = 0.496$, $p < 0.001$), followed by ethical trust ($\beta = 0.454$, $p < 0.001$) and functional trust ($\beta = 0.258$, $p < 0.01$). Also, citizen trust has a significant influence on service acceptance ($\beta = 0.777$, $p < 0.001$). These results indicate that governance mechanisms, ethical safeguards, and operational reliability collectively contribute to trust-building.

Table 11. Hypothesis testing using bootstrapping**.**

Path	β	t-value	p-value	Decision
ET → CT	0.454	4.217	<0.001	Supported
FT → CT	0.258	2.631	0.009	Supported
IT → CT	0.496	4.893	<0.001	Supported
CT → SA	0.777	11.824	<0.001	Supported

To assess the practical applicability of the proposed model, three leading smart-city initiatives, i.e., Barcelona, Amsterdam, and Helsinki, are examined. Barcelona demonstrates the ethical trust layer through its citizen-centric data governance and digital sovereignty initiatives, which improve privacy protections and citizens' control over their personal data. Helsinki demonstrates both ethical and functional trust through its explainable AI and transparent public algorithms. Amsterdam reflects the functional trust layer by including participatory governance, human oversight, and citizen engagement mechanisms in AI deployment. These findings furnish real-world support for the proposed model and indicate that ethical trust, functional trust, and institutional trust collectively promote sustainable public trust in 5G-enabled smart-city ecosystems, as demonstrated in Table 12.

Table 12. Case study validation.

City	Dominant Layer	Key HCAI Practices	Trust Outcome
Barcelona	Ethical Trust	Digital sovereignty, privacy shield, transparent data governance	Improved citizen trust and data control
Helsinki	Ethical and Functional Trust	Explainable AI, algorithmic transparency	Enhanced trust in AI-enabled public services
Amsterdam	Functional Trust	Human oversight, citizen participation,	Improved transparency and public trust

We have conducted a bibliometric analysis based on the selected studies using R Studio. Fig. 3 shows the results of a word cloud analysis using keywords from the selected papers to identify the most dominant keywords in this study area. The smart city illustrates the most familiar occurrence, observed by human-centric, internet of things, urban planning, security and privacy, emerging technologies, human behavior, and automation.

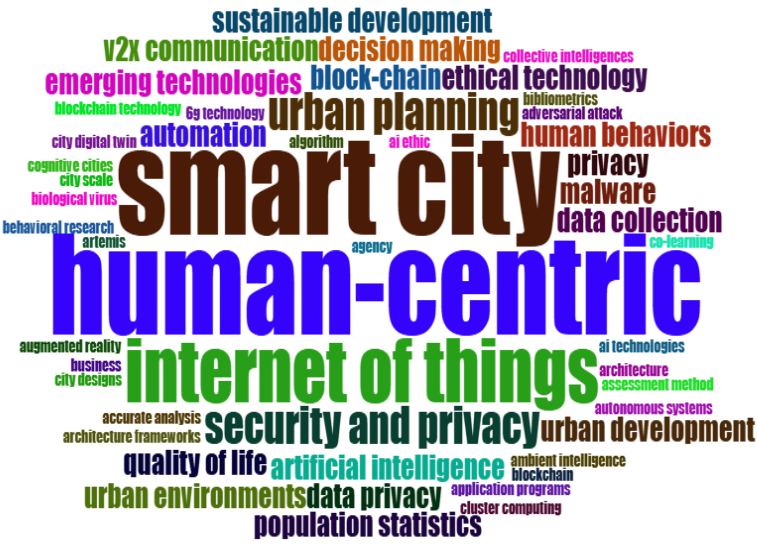

Fig. 3. Word cloud analysis based on the identified studies.

Fig. 4 displays the network cluster analysis of the identified studies, with three clusters recognized in red, blue, and green. The red and blue clusters are context clusters that group keywords related to smart cities, data privacy, decision-making, and AI tools.

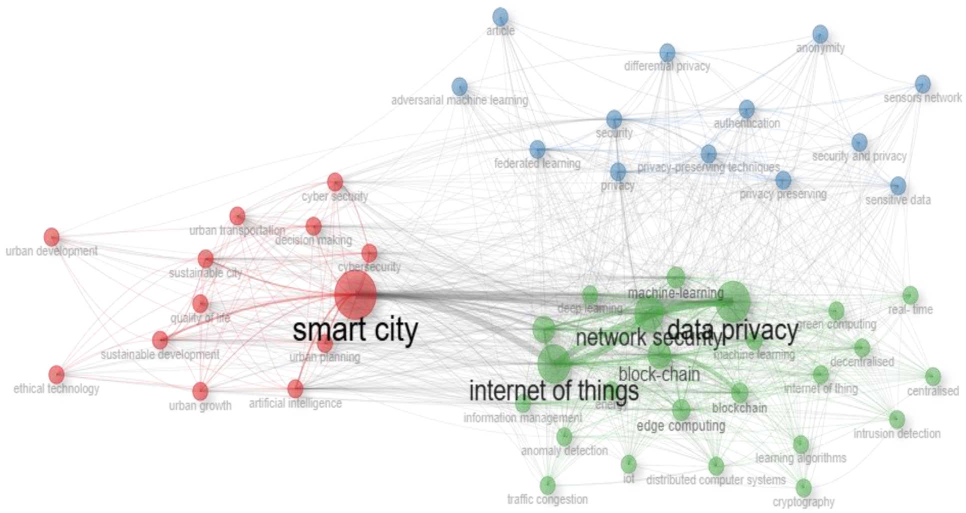

Fig. 4. Network cluster analysis.